

Abdullah Al Zahid

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SUMMARY

Transportation engineer with a master's degree whose work has centred on simulation: agent-based (MATSim) and macroscopic network models in graduate research, traffic simulation of Ontario Line construction staging, and OpenTrack rail simulation of GO services at Metrolinx. Builds the Python tooling behind the analysis and turns model output into clear reports for decision-makers.

EXPERIENCE

Assistant Planner, Transportation Planning

City of Toronto | January 2024 – present

- Analyze travel and network data, including several years of the Transportation Tomorrow Survey, to establish where capacity falls short and what that means for future service and investment.
- Built the Transportation Data Dashboard, replacing a report rebuilt by hand each quarter with one validated source the division reports from, and saw the change through to adoption.
- Review development applications and the transportation studies filed with them against City standards and Official Plan policy, and recommend conditions where something is missing.
- Write the technical reports, briefing notes and presentations that carry findings to planners, engineers and managers across divisions.

Transportation Planner, Simulation and Modelling

Metrolinx, Service Design | February 2023 – January 2024

- Ran OpenTrack simulations of GO rail services to build and test timetables against infrastructure constraints, speed restrictions and corridor conditions.
- Validated proposed operating changes by establishing what they would do to run times, schedule reliability and recovery before anyone committed to them.
- Built the Crew Variance Tool, a Python application that read large JSON operations files and produced summaries of speed profiles and run-time variance across subdivisions.
- Set up a command line workflow so temporary and permanent slow orders could be pre-simulated in batches, cutting turnaround on scenario requests from days to hours.
- Worked with schedulers, operations staff and engineers to settle the variables a service plan had to be built on.

Graduate Transportation Planner

Mott MacDonald | January – February 2023

- Short consulting placement on Ontario Line work. Ran traffic simulations to estimate how construction staging would affect surrounding streets, and prepared the supporting material for the project team.

Graduate Research Assistant

University of Alberta | January 2020 – August 2022

- Master's research on how road networks perform during emergency evacuations, using macroscopic (LWR) and agent-based (MATSim) models.
- Worked with large travel datasets and wrote up results for readers without a modelling background.

SKILLS

Agent-based and microscopic simulation: MATSim, VISSIM, Synchro and OpenTrack. Python (Pandas, NumPy) and SQL for analysis and automation. Model testing and data visualization. Advanced Excel. ArcGIS and QGIS. Technical reports and presentations.

EDUCATION

MSc, Transportation Engineering — University of Alberta, 2022

BSc, Civil Engineering — Bangladesh University of Engineering and Technology, 2018

BSc, Computer Science — University of British Columbia, in progress (2025–2027)